

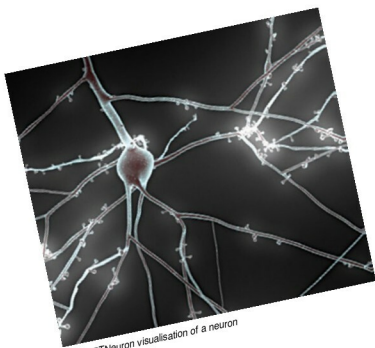


Fig. 2: RTNeuron visualisation of a neuron

The simulator neuron is used with automatically fitted algorithms running on Blue Gene to insert ion channels and adjust their parameters to capture specific electrical properties of the different electrical types found in each anatomical class. Three-dimensional neurons are placed in the layers of the Blue builder, which is a circuit builder. Axo-dendritic touches are maintained and analysed by an algorithm known as collision algorithm.

Modelling and simulating the microcircuit

In 1952, Hodgkin and Huxley published the highly-successful model of ionic currents that allowed simulation of the action potential. These simulations show the behaviour of an ion channel. This ion channel can be used in an artificial neuron for transferring the action potential. Neurons are simulated into microcircuits, which are embedded on a local circuit attached in the brain. From there, whole brains are simulated.

Number of neurons are attached to a single neuron for efficient data

transferring. Neurons will receive data and transfer it into thousands of other neurons, so the microcircuit must be small and less complex, but it must fully sustain the neurons. A massive increase in computational power is required to make this quantum leap—an increase that is provided by IBM's Blue Gene super-computer.

The neocortical column is the basic functional unit of the cerebral cortex in mammals. These functional units are in millions in the cortex, which are similar in rats and humans. The laboratory has prepared for this reconstruction by developing the multi-neuron patch-clamp approach, recording from thousands of neocortical neurons and their synaptic connections. A four- to sixteen-day-old rat's somato sensory cortex is used for recording, as visualisation of their tissues is optimal at this age.

The template, Blue column, will be composed of ~10,000 neocortical neurons within the dimensions of a neocortical column (~0.5mm in diameter and ~1.5mm in height).

Fig. 1 shows the neocortical column (NCC) microcircuit in various stages of reconstruction. These are arranged in layers. However, in this figure, only a fraction is reconstructed. Neurons are shown in 3D; red-coloured parts indicate dendrite and blue-coloured parts indicate axonal arborisation.

When the microcircuit is built, the circuit is started. For this, the power of Blue Gene computer is used. All the 8192 processors of the computer are pressed into service. The process is highly parallel so it can easily solve highly-complex computation. As the electrical impulse travels from neuron to neuron, results are communicated via inter-13 processor communication (MPI).

RTNeuron

RTNeuron software, developed for Blue Brain project, is used for visualisation of the neural simulation.

This software is specific to neural simulation. It is written in C++ and OpenGL language. It can accept the simulation output and render them into a 3D model so researchers can analyse the neural structure. Propagations of action potential are visualised so scientists can pause, zoom and analyse them.

Uploading the human brain

One of the major advantages of the virtual brain project is its ability to upload human brain into the machine and keeping alive the knowledge of the human after his or her death. It can also remember things such as phone numbers, important events, historical facts, etc, without any effort. It can also take decisions and make calculations with ease.

Nanobots are used here for scanning the human brain. These nanobots are always connected to the computer and are small enough to travel through the circulatory system. These can travel to the spine and brain, scan the human brain, analyse its structure and send data into the computer for further calculation. ●